

# In the Future Metaverse, What Kind of UGC do Users Need?

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## ABSTRACT

With the COVID-19 pandemic, people's real-life interactions diminished, and the game-based metaverse platforms such as *Minecraft* and *Roblox* are on the rise. The main users of these platforms are teenagers, they generate content in a virtual environment, which can significantly increase the activity of the platform. However, the experience of User-Generated Content in the metaverse is not very good. So what kind of support do users need to improve the efficiency of generating content in the metaverse? To investigate teenage users' preferences and expectations of it, this paper interviewed 72 teenagers aged 12-22 who are familiar with the metaverse game, and distilled 4 suggestions that can help promote metaverse users to generate content.

**Index Terms:** Metaverse—User-Generated Content—Game—Artificial intelligence

## 1 INTRODUCTION

The metaverse first appeared in 1992 in Neal Stephenson's science fiction novel *Snow Crash*. It refers to an Internet-based 3D virtual world<sup>[1]</sup> that allows people to live and work in an immersive 3D virtual world through virtual characters<sup>[2]</sup>. Kaiyuan Securities Co., Ltd in China believe that open world games, mainly *Minecraft* and *Roblox*, have preliminary metaverse elements and may represent the development trend of the game form<sup>[3]</sup>. *Roblox*, for example, has 150 million monthly active users (MAU) in 2022 and is used by 2/3 of 9-12 year olds in the US, with 1/3 of them under 16 years old. In addition, 72% of the top 1000 creators on *Roblox* are under the age of 13<sup>[4]</sup>. It is evident that younger users can be well engaged in the metaverse content creation.

User-generated content (UGC) is any form of content created by users rather than developers of online platforms<sup>[5]</sup>. It can substantially increase the efficiency and richness of content creation on metaverse platforms and extend the platform lifecycle<sup>[3]</sup>. The development of the metaverse needs to rely on user-generated digital content; therefore, this study intends to explore what kind of UGC support users need from the platform by interviewing teenage metaverse users, and to provide suggestions for improving the efficiency of User-generated content.

## 2 RELATED WORK

At present, there are fewer studies discussing User Generated Content in game-based metaverse platforms. Haihan Duan reviewed (2022) representative UGC in video games, and listed the methods of UGC propagation in video games<sup>[6]</sup>. Daji Shin and

Jae-young Yoon derived a paradigm model of game-based metaverse user experience<sup>[7]</sup>. Qiming Po found that the premise of producing UGC in the Metaverse to share or sell profitably with the platform is the close integration of physical strength and intelligence with key virtual production<sup>[8]</sup>. In this paper, we focus on User Generated Content in the metaverse and explore what kind of support mechanisms users need.

## 3 METHODOLOGY

### 3.1 Research Methodology

In this study, we used the interview method and designed the interview questions as an open-ended questionnaire that could be filled out online. For the interview data, we used NVivo12 for open-ended coding and spindle coding to sort out the respondents' views on metaverse UGC. The interviews were stopped only when they reached saturation and confirmed that there were no new findings.

### 3.2 Sample collection

This study focused on interviewing metaverse users aged 12-22 living in China, who at least have played one metaverse game for more than 3 months (e.g., *Minecraft*, *Roblox*, or *Fortnite*) and had engaged in User Generated Content.

When collecting interview materials of those who are 12-18 years old, we search for them in metaverse game communities such as *Minecraft* and *Roblox*, as well as the social platforms such as *Weibo* and *Douban*. We entered their online communities, sent the questionnaire's link to communities for them to fill out online. Finally, we received a total of 55 responses, and screened out 30 responses which filled out by teenagers aged between 12 and 18. Most of whom were in elementary, middle or high school.

When collecting interview materials from the 18-22 year olds, since people of this age are normally studying for their undergraduate or master's degree, our researchers selected four elective courses related to metaverse at the authors' university in September to October 2022, entered the classroom to observe student-teacher interactions, screened 100 students who were interested in metaverse UGC, and distributed questionnaires to them via *WeChat*. After screening, 42 responses were selected as valid interview materials.

In summary, there are 72 valid interview materials collected in this study, including 22 females and 50 males. Participants were not paid.

### 3.3 Interview content

The interview was divided into two phases.

The first phase focused on three questions: what kind of content users would like to create in the metaverse; what kind of features users expect about User-Generated Content, and what kind of features would help improve the efficiency of User-Generated Content in the metaverse.

In the second phase, the interview outline was rewritten based on the above results. Using the online questionnaire, we conducted a second interview, and deeply refine the findings based on the interviews.

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#### 4 FINDING

After coding, we found that respondents had some common expectations.

The coding results are as follows.

|   | Code                                                                                                                           | The frequency of being noticed |
|---|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 1 | Interaction function                                                                                                           | 30.0%                          |
| 2 | Multi-player cooperation function                                                                                              | 27.5%                          |
| 3 | Multidimensional sensory and emotional communication functions                                                                 | 24.2%                          |
| 4 | Simulated reality functions                                                                                                    | 12.5%                          |
| 5 | Other Functions<br>(Realization function, multi-platform link function, information retrieval function, reproduction function) | 5.8%                           |

Table 1: Evaluation of Case Study (n=72)

#### 5 DISCUSSION

Based on the previous findings, this study summarizes 4 suggestions that are conducive to improve the efficiency of UGC.

##### 5.1 Collaborative multi-person creation

In an ideal metaverse, collaborative multi-person creation can quickly turn concepts into results. This mechanism is manifested by team building, real-time communication, and online operation. When creating content in the metaverse, we may have misunderstandings with teammates due to differences in individual perceptions. The function needs to be more collaborative, making knowledge tangible, and allowing people to explain idea with less bias caused by verbal representation. In addition, it is also necessary to establish a perfect progress saving and progress regression function, and pay more attention to protect user privacy when creating content in multi-player collaboration.

##### 5.2 Intelligent Interaction Scene Creation

According to the interviews, it is necessary to use intelligent interaction scenes to assist metaverse content production. First, the mechanism is capable of recognizing and responding to voice interaction, body interaction, and gesture interaction. By adding body sensing function, users can complete a series of actions in a more physical way. Second, the mechanism can realize personalization, predict the user's next action or choice, and realize interactive, programmable, personalized rather than formatted multi-possibility creation. Third, the mechanism enables a high degree of freedom and an infinite number of details in the scene.

##### 5.3 Collaborative creation with AI

During the interviews, we found that AI was mentioned frequently. Users expect to be able to use AI to assist in the creation of models and characters, plots, scenes, and even to use AI to assist in decision making and fill in gaps. This mechanism has the following two characteristics: automatic generation and style fusion. The first is auto-generation. Using AI to generate distant views of the virtual world to deceive the human, could help to reduce the amount of computation required to simulate the entire virtual world. Moreover, combining AI with human subjectivity can achieve high efficiency in creation. The second is the fusion of styles. AI finds the best combination of various hybrids of

works and can provide new inspiration for User Generated Content.

##### 5.4 Improve the conversion mechanism of economic and social benefits

In the future, the economic benefits of metaverse UGC can come from the sale of digital rights or from the metaverse e-commerce platform, and users can reproduce the services of travel, fitness, education, etc. perceptually. The metaverse should at least cooperate with real-life e-commerce companies to provide a complete metaverse backup of goods, and allow the sale and printing of works.

In terms of social benefits, metaverse content production can be applied to educational activities or social events, such as simulating economic crises and other phenomena that cannot be simulated in the real world.

Of course, some users in the interview expressed different views, they think the benefits are a double-edged sword and do not want metaverse UGC to generate large-scale economic or social benefits.

#### 6 CONCLUSION

In the above discussion, we came up with some mechanisms as suggestions. These suggestions are very necessary and important, pointing out the direction of metaverse User Generated Content. These mechanisms have requirements for computational power and fidelity of simulation. We need excellent hardware support, because the current smart interaction hardware is feature-rich but not light enough. In the future, we should focus on discovering interactive devices that are lighter than today's VR glasses and build more sensitive brain-computer interfaces to achieve multi-dimensional sensory simulation and sensory action reading, such as the display of facial expressions, scene selection, image construction, and so on.

We have enough reason to believe that in the near future, the system of metaverse User Generated Content will be more perfect, providing a powerful boost to the development of metaverse and even the real world.

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